

Visual-Inertial SLAM with Extended Kalman Filter for Stereo-IMU Fusion

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Abstract—This project implements a visual-inertial simultaneous localization and mapping (SLAM) system using an extended Kalman filter (EKF). The inertial part performs pose prediction on the Lie group $SE(3)$ using linear velocity and angular velocity measurements. For the visual part, stereo observations are modeled to update both the pose and a set of landmark positions. To maintain numerical stability, the implementation uses innovation gating (pixel residual and NIS threshold) and covariance regularization to avoid invalid/indefinite matrices. Experiments on provided datasets (00, 01, and 02) show that the SLAM trajectory is primarily driven by IMU prediction while being corrected by the estimated landmarks.

Index Terms—Visual-inertial SLAM, extended Kalman filter, stereo vision, $SE(3)$ pose propagation, landmark mapping.

I. INTRODUCTION

Visual-inertial SLAM combines motion information from an inertial measurement unit (IMU) with geometric constraints from a camera to estimate the trajectory of a moving sensor and the locations of scene landmarks. Compared with using only a monocular/stereo vision pipeline, adding IMU measurements helps reduce scale and motion ambiguity and improves robustness during periods of weak visual constraints. Compared with pure dead-reckoning from IMU, camera measurements correct drift over time.

Approach Overview

The implementation follows an EKF structure: First, an $SE(3)$ -based EKF prediction step propagates the IMU pose and covariance using the measured twist. Second, stereo feature observations are used to perform EKF updates for both the pose and a selected subset of landmark positions. Landmarks are initialized via stereo triangulation and are updated as new measurements become available. To control computation, only a bounded number of landmarks are kept in the filter state.

II. PROBLEM FORMULATION

Let $T_t \in SE(3)$ denote the IMU pose at time t , represented as a homogeneous transform. We model scene landmarks as a set of static points $m_i \in \mathbb{R}^3$ for $i = 1, \dots, M$.

Inputs and Measurements

At each time step, the IMU provides linear velocity $v_t \in \mathbb{R}^3$ and angular velocity $\omega_t \in \mathbb{R}^3$ (both expressed in the IMU/body frame). Stereo camera observations are provided as pixel coordinates

$$z_{t,i} = \begin{bmatrix} l_{x,t,i} \\ l_{y,t,i} \\ r_{x,t,i} \\ r_{y,t,i} \end{bmatrix} \in \mathbb{R}^4, \quad (1)$$

or $z_{t,i} = [-1, -1, -1, -1]^\top$ if landmark i is not observed at time t .

Desired Output

The goal is to estimate:

- the pose trajectory $\{T_t\}$,
- the landmark positions $\{m_i\}$.

Observation Model

Given a pose T_t and a landmark m_i , the left and right camera coordinates are obtained from known extrinsic calibration matrices. A pinhole projection yields predicted stereo pixel coordinates $\hat{z}_{t,i} = h(T_t, m_i)$. The EKF update uses the innovation

$$\text{inn}_{t,i} = z_{t,i} - \hat{z}_{t,i}. \quad (2)$$

III. TECHNICAL APPROACH

A. Pose Representation and Prediction on $SE(3)$

The pose is propagated using $SE(3)$ kinematics. Let $u_t = [v_t^\top, \omega_t^\top]^\top \in \mathbb{R}^6$ be the measured twist. For a time increment Δt , we compute the incremental motion via the exponential map:

$$T_{t+1} = T_t \exp(\widehat{u_t \Delta t}). \quad (3)$$

In the implementation, the 6D twist is mapped to a 4×4 transform using an `axangle2pose` utility.

B. EKF State and Covariance Propagation

The EKF state couples:

- a 6-DOF pose perturbation (local coordinates),
- 3M landmark coordinates.

Prediction updates the pose covariance block using the adjoint representation (`pose2adpose`) and adds continuous-time process noise scaled by Δt .

C. Landmark Mapping (Part b)

Given a fixed (predicted) IMU trajectory, landmark positions are estimated by an EKF that maintains a mean μ_m and covariance σ_m for each landmark. For a set of observed landmarks at time t , the stereo projection Jacobians define a measurement Jacobian H . The update follows the standard EKF form:

$$S = H\sigma H^\top + V, \quad (4)$$

$$K = \sigma H^\top S^{-1}, \quad (5)$$

$$\mu \leftarrow \mu + K \text{inn}, \quad (6)$$

$$\sigma \leftarrow (I - KH)\sigma. \quad (7)$$

Triangulated initialization uses stereo observations and discards points that violate simple depth and reprojection error checks.

D. Visual-Inertial SLAM (Part c)

The full SLAM EKF estimates both the pose and selected landmarks. At each time step:

Feature Selection: Only a subset of features is kept in the filter to control the computational cost. The implementation selects at most `max_features` points by dividing the timeline into windows and preferring features with higher observation counts.

Landmark Initialization: When a feature is observed for the first time, its initial 3D position is obtained via stereo triangulation (DLT-based in the utilities). Initialization is accepted only if the landmark satisfies:

- positive depth in the IMU-related camera frame,
- reprojection residual below `init_reproj_thresh`.

Measurement Update and Jacobians: For each candidate landmark, the EKF computes:

- predicted stereo projection $\hat{z}$,
- innovation inn ,
- Jacobians with respect to pose and landmark coordinates.

The total measurement Jacobian H_c is built from the pose and landmark blocks and used to form the innovation covariance:

$$S = H_c \sigma_{cc} H_c^\top + V, \quad (8)$$

where σ_{cc} is the local sub-block of the full covariance for the pose and the current landmark.

Gating and Numerical Stability: To avoid spurious updates:

- Pixel residual gating rejects large innovations: $\|\text{inn}\|_\infty \leq \text{pixel_gate}$.
- NIS gating rejects inconsistent measurements: $\text{NIS} = \text{inn}^\top S^{-1} \text{inn}$ with a χ^2 -based threshold.
- Innovation covariance S and local covariance blocks are symmetrized and eigenvalue-regularized to improve positive definiteness.

The filter also bounds the correction magnitude to reduce trajectory collapse (both translation and rotation corrections are clipped).

State Update: The pose mean is updated on $SE(3)$ using the exponential map:

$$T_t \leftarrow T_t \exp(\hat{\delta\xi}). \quad (9)$$

Landmarks are updated additively:

$$m_i \leftarrow m_i + \delta m_i. \quad (10)$$

E. IMU Bias Handling

The implementation removes a constant bias component by subtracting the mean of expected near-zero axes: the vertical component of v_t and the horizontal components of ω_t . This heuristic improves the quality of the IMU-only baseline and reduces systematic drift.

IV. RESULTS

This section reports qualitative results across datasets. All plots were generated from the provided dataset sequences using the same pipeline.

A. Dataset 00

Figure 1 compares IMU-only localization, landmark mapping, and full visual-inertial SLAM. The SLAM trajectory is corrected using the estimated landmarks rather than following the IMU trajectory alone.

B. Dataset 01

Figure 2 shows similar comparisons on dataset 01.

C. Dataset 02

Figure 3 shows similar comparisons on dataset 02. In this dataset, the gating and the number of EKF updates per step are tuned to maintain filter stability while still allowing visual correction.

D. Discussion: What Worked and What Did Not

In practice, several issues can cause the SLAM EKF to produce unstable or overly constrained trajectories:

- If landmark initialization accepts poor points, early incorrect landmarks can bias later pose updates.
- If the measurement noise or gating is too permissive, inconsistent feature correspondences can over-correct the pose.
- If gating is too strict, the filter becomes effectively IMU-only because too few observations pass.

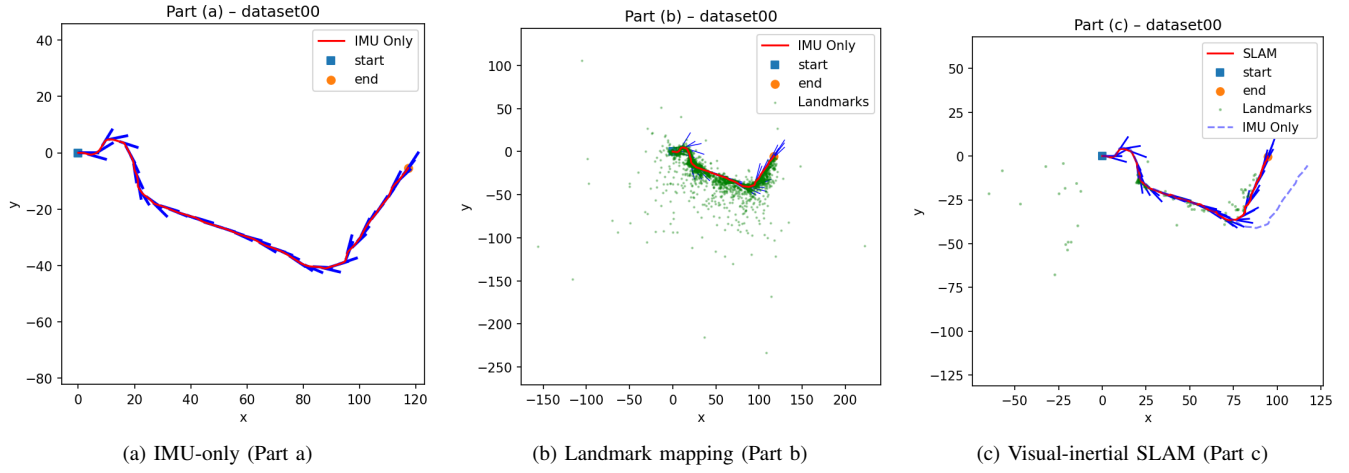


Fig. 1. Dataset 00 results.

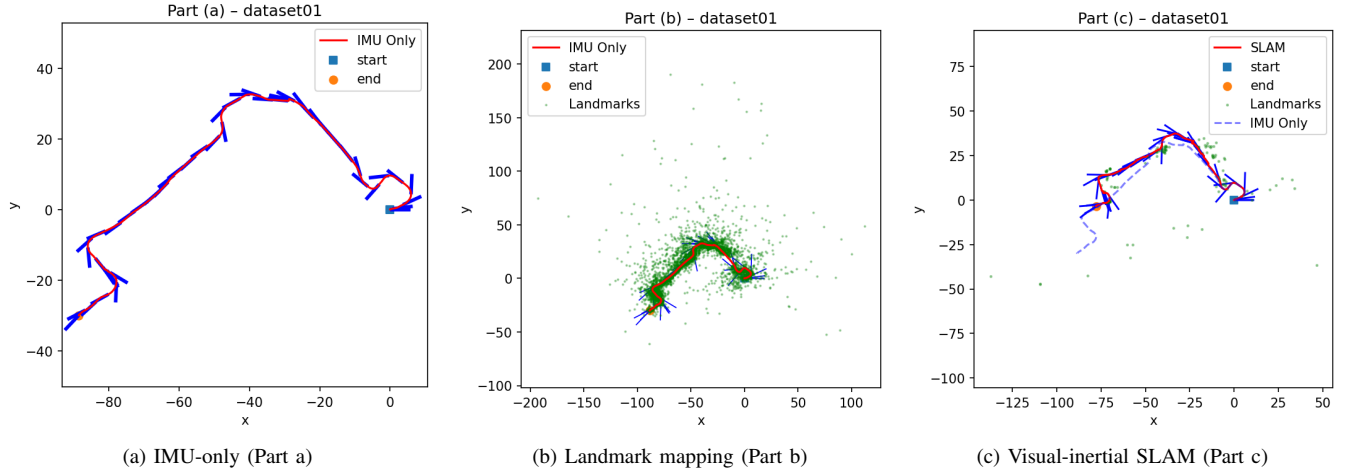


Fig. 2. Dataset 01 results.

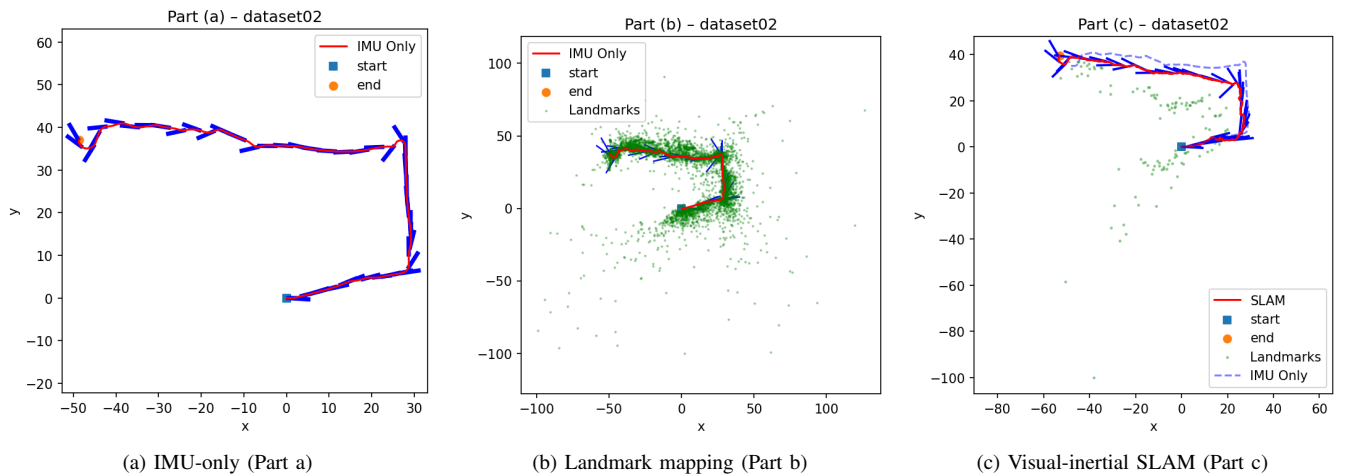


Fig. 3. Dataset 02 results.

To address these failure modes, the implementation uses:

- reprojection/depth checks during landmark initialization,
- pixel residual and NIS gating during updates,
- covariance symmetrization and eigenvalue regularization.

Overall, these techniques improved numerical stability and allowed the SLAM trajectory to be visually aligned with the estimated landmark set.

V. CONCLUSION

This project implements a visual-inertial EKF-SLAM system on $SE(3)$ for pose prediction and stereo projection updates for landmark estimation. The key contributions are: (i) Lie-group prediction with covariance propagation on $SE(3)$, (ii) stereo observation modeling with EKF updates for pose and landmarks, (iii) measurement gating and covariance regularization to ensure stable filter behavior. Future improvements could include more principled adaptive noise tuning and more robust feature/landmark management to reduce the need for dataset-specific parameters.

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